



BULK PO HEADER UPDATE USING SOAP WEB SERVICE IN EXCEL VBA

Abstract

This solution provides a practical method to mass-update Purchase Order (PO) Header data in Oracle Fusion Cloud using SOAP Web Services automated through an Excel VBA tool. It allows organizations to efficiently modify header attributes—including DFFs—without affecting PO lines. The process uses a customizable XML payload, supports bulk updates, and includes validation, security setup, and error handling. While updates are restricted for POs that are finally closed or have progress payments enabled, the method remains an effective and reliable approach for large-scale PO header corrections.

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1. Purpose

This document describes the complete end-to-end process for mass-updating Purchase Order (PO) Header information in Oracle Fusion Cloud by using:

- **SOAP Web Services**
- **Excel with a VBA macro**
- **A customizable XML payload**

This solution allows you to update any updatable PO Header field, including but not limited to:

- **Payment Terms**
- **Buyer**
- **Bill-to, Ship-to, Location fields**
- **Supplier or Supplier Site (only if allowed)**
- **Header-level attributes**
- **Descriptive Flexfields (DFFs)**
- **Any standard header element supported by the PurchaseOrderServiceV2 API**

The method does not modify PO lines, schedules, or distributions, making it safe for large-scale administrative corrections.

2. Scope

This method applies to:

- **Oracle Fusion Cloud Procurement**
- **Any environment (TEST / PROD)**
- **Any PO header attribute supported by the SOAP service**
- **Bulk or single-PO updates**

Exclusions / Limitations

Functional Limitations

- **POs with Progress Payments enabled cannot be updated**
Oracle enforces this restriction across SOAP .
- **POs assigned to inactive Buyer accounts cannot be updated**
The **Buyer is always treated as the Initiator of the Change Order**.
If the Buyer is end-dated, terminated, or missing Procurement roles, the Change Order cannot be created.
- **Cancelled, Closed, or Finally Closed POs cannot be updated**
These documents are treated as historical and cannot accept Change Orders.
- **Header fields restricted by workflow or approvals may reject updates**
If organizational rules require approvals or block changes, the service will enforce those rules.

Technical / Operational Limitations

- **Bulk Updates Cause Expected Excel Lag**
When processing large datasets, Excel will appear to freeze or hang.
This is normal because:
 - Each row triggers a SOAP request
 - Fusion processes each request and creates Change Orders in sequence
 - Excel cannot repaint the screen during heavy looping

The macro must be left running until it completes.

Do not close Excel during this period — the process is still working normally.

- **Fusion UI may temporarily slow down**
Large volumes of Change Orders may cause:
 - Slower Procurement screens
 - Delayed Change Order visibility
 - Temporary performance dips

All updates continue to process successfully in the background.

- **DFF value sets must allow the value being assigned**
If the value is not allowed in the LOV, Oracle will reject it with:
FND_VS_VAL_NOT_IN_LIST

3. Supported Header Attributes

The web service supports updates to **any element defined in the PurchaseOrderServiceV2 header structure**, including:

✓ Standard Header Fields

- PaymentTermsId
- BuyerId
- BillToLocationId
- ShipToLocationId
- SupplierId / SupplierSiteId (with restrictions)
- FOB
- FreightTerms
- CurrencyCode
- CarrierId
- NoteToReceiver
- NoteToVendor

✓ DFF Segments

- Any segment defined in **Purchasing Document Header** flexfield
- Update via <HeaderFlexfield> element

✓ Other Header Elements

- Change order description

- Approval flags (e.g., BYPASS)
- Any accessible attribute exposed in the service schema

This makes the solution extremely flexible and reusable.

4. Pre-Update Requirements

4.1 DFF or Attribute Deployment (if applicable)

If updating DFFs:

- Ensure the **Purchasing Document Header** flexfield is deployed
- Confirm the value set contains allowed values
- Validate the segment API name

4.2 Buyer Status (Critical)

The **Buyer assigned to the PO must be active**:

- Buyer must have an active HCM person record
- Buyer must not be terminated or end-dated
- Buyer must still hold procurement roles

If the Buyer is inactive, the system will prevent the creation of a Change Order.

4.3 Required User Privileges

Privilege	Purpose
Manage Purchase Order by Web Service	Allows editing via SOAP
Search Purchase Order by Web Service	Allows querying POs
Bypass Purchasing Document Approvals	Optional; auto-approval

Best practice: Create a **custom integration role** containing these privileges.

Custom_PurchaseOrder_WebService_Role

Custom_PurchaseOrder_WebService_Role

Procurement - Abstract Roles

Enable Permission Groups

Custom role used for SOAP-based Purchase Order header updates including DFF modifications.

Privileges Resources

[+ Add Function Security Policy](#) [✕ Delete](#) [📄](#) [📄 Detach](#)

Privilege Name	Inherited from Role	Description
Import Purchasing Document Bypassing Approval		Allows a procurement agent to bypass approvals when importing purchase orders or purchase agreements and when creating purchase orders through the purchase order web service.
Manage Purchase Order by Web Service		Allows management of purchase orders using web service.
Search Purchase Order by Web Service		Allows searching for purchase orders using the web service.

Load Inherited Policies

Details

 Detach

Resource Name	Description
No data to display.	

5. Web Service Endpoint

All updates use:

/fscmService/PurchaseOrderServiceV2

Full example:

https://your_instance_link.oraclecloud.com/fscmService/PurchaseOrderServiceV2

6. Excel Workbook Structure

Sheet: DATA

Column Purpose

A	PO Header ID
B	PO Number (SEGMENT1)
C	Optional dynamic attributes (DFF or standard fields)
D	Generated payload
E	Web service response

	A	B	C	D	E
1	PO Header Id	SEGMENT1	ATTRIBUTE13	Payload	Return
2	300000015733908	PO-24-00040			
3	15010	PO-00158			
4	7016	PO-00080			
5	7011	PO-00070			
6	300000009251719	PO-23-00021			
7	300000009251748	PO-23-00025			
8	300000009251758	PO-23-00026			
9	300000009275358	PO-23-00030			
10	300000009275422	PO-23-00031			
11	300000009275512	PO-23-00032			
12	300000009275524	PO-23-00033			
13	300000009275613	PO-23-00034			
14	300000009321125	PO-23-00035			
15	300000009321409	PO-23-00036			
16	300000009355201	PO-23-00037			
17	300000009355214	PO-23-00038			
18	300000009355322	PO-23-00039			
19	300000009355466	PO-23-00040			
20	300000009355472	PO-23-00041			

7. Generic SOAP Payload Template

(Updated to include instructions for DFF API names and forced values)

This payload is flexible and can update any standard PO header attribute.

If you are updating DFFs, you must use the API Name of the flexfield segment in the tag and insert the desired value directly.

✓ Example for updating a DFF

If your DFF segment API name is prType, then the XML must include:

`<ns3:prType>Development</ns3:prType>`

Where:

- prType → API Name of the DFF segment (not the display label)
- Development → The value you want to force on the PO

Sheet: PAYLOAD_TEMPLATE

Cell A1 stores the XML payload template.

A1	<code><soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/"</code>
	<code>xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"</code>
	<code>xmlns:xsd="http://www.w3.org/2001/XMLSchema"></code>
	<code><soap:Body></code>
	<code><ns1:changePurchaseOrder</code>
	<code>xmlns:ns1="http://xmlns.oracle.com/apps/prc/po/editDocument/purchaseOrderServiceV2/types/"</code>
	<code><ns1:changeOrderEntry xmlns:ns2="http://xmlns.oracle.com/apps/prc/po/editDocument/purchaseOrderServiceV2/"</code>
	<code><ns2:POHeaderId>XXXXXXXXXX</ns2:POHeaderId></code>
	<code><ns2:OrderNumber>YYYYYYYYYY</ns2:OrderNumber></code>
	<code><ns2:ChangeOrderDescription>To update pr type as requested from procurement team</ns2:ChangeOrderDescription></code>
	<code><ns2:ApprovalActionCode>BYPASS</ns2:ApprovalActionCode></code>
	<code><ns2:HeaderFlexfield</code>
	<code>xmlns:ns3="http://xmlns.oracle.com/apps/prc/po/editDocument/flex/draftPurchasingDocumentHeader/"</code>
	<code><ns3:prType>Development</ns3:prType></code>
	<code></ns2:HeaderFlexfield></code>
	<code><ns2:Action>CHANGE</ns2:Action></code>
	<code></ns1:changeOrderEntry></code>
	<code></ns1:changePurchaseOrder></code>
	<code></soap:Body></code>
1	<code></soap:Envelope></code>
2	

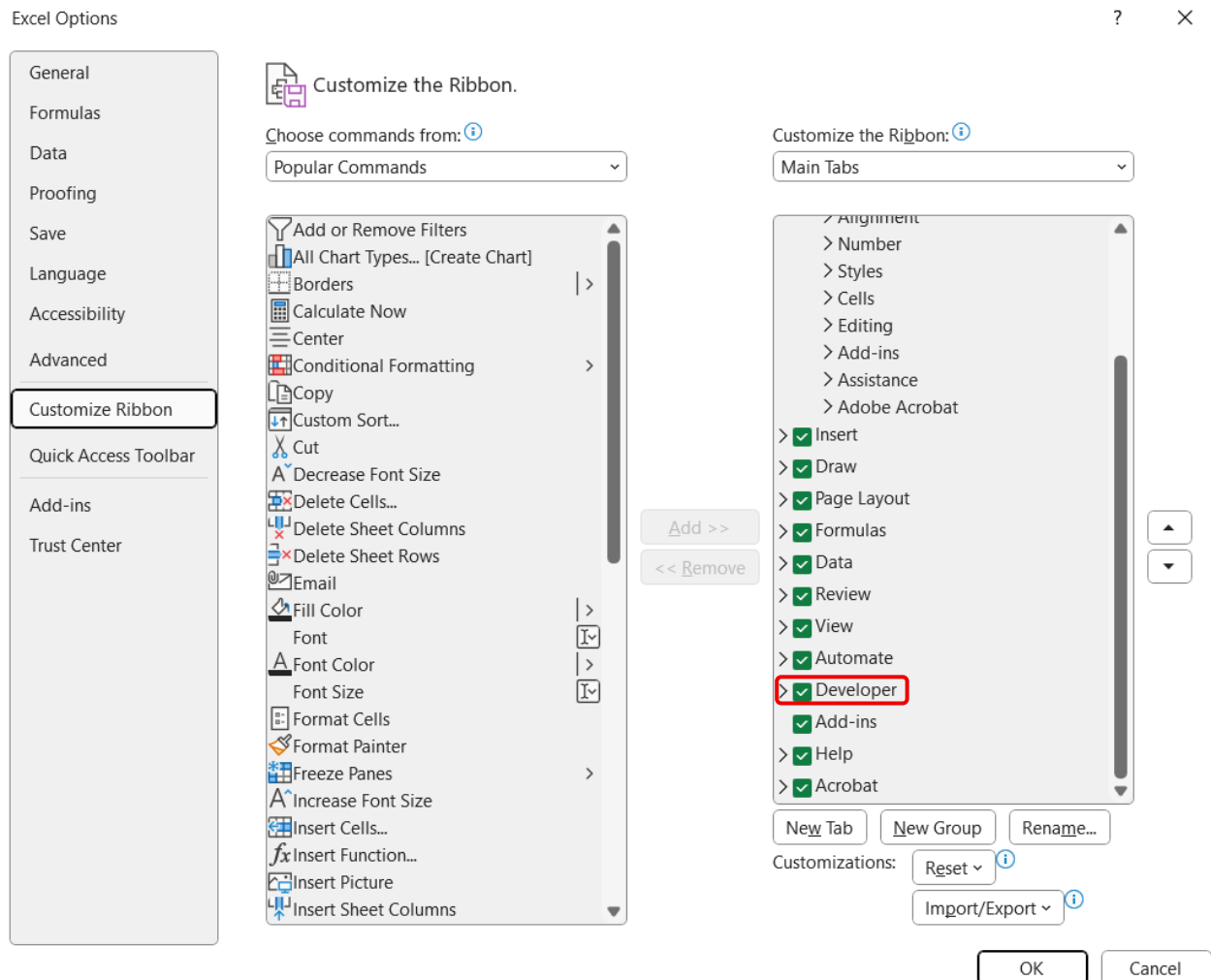
8. Excel Prerequisites (Mandatory Before Running the Macro)

To allow Excel to run VBA code and call APIs, you must enable specific trust settings.

8.1 Enable the Developer Tab

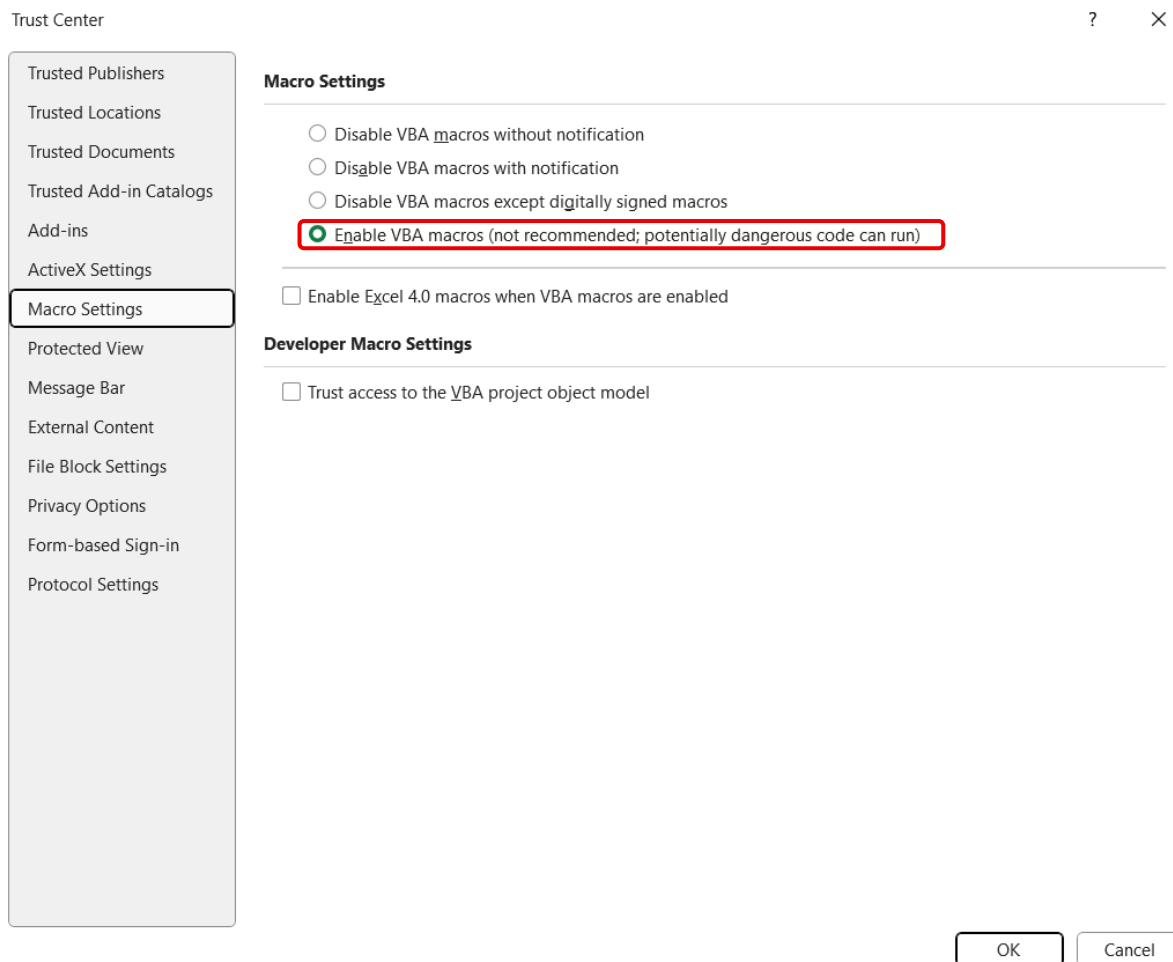
1. Go to **File → Options**
2. Select **Customize Ribbon**
3. Check **Developer**
4. Click **OK**

You will now see the **Developer** tab in your Ribbon.



8.2 Enable Macros

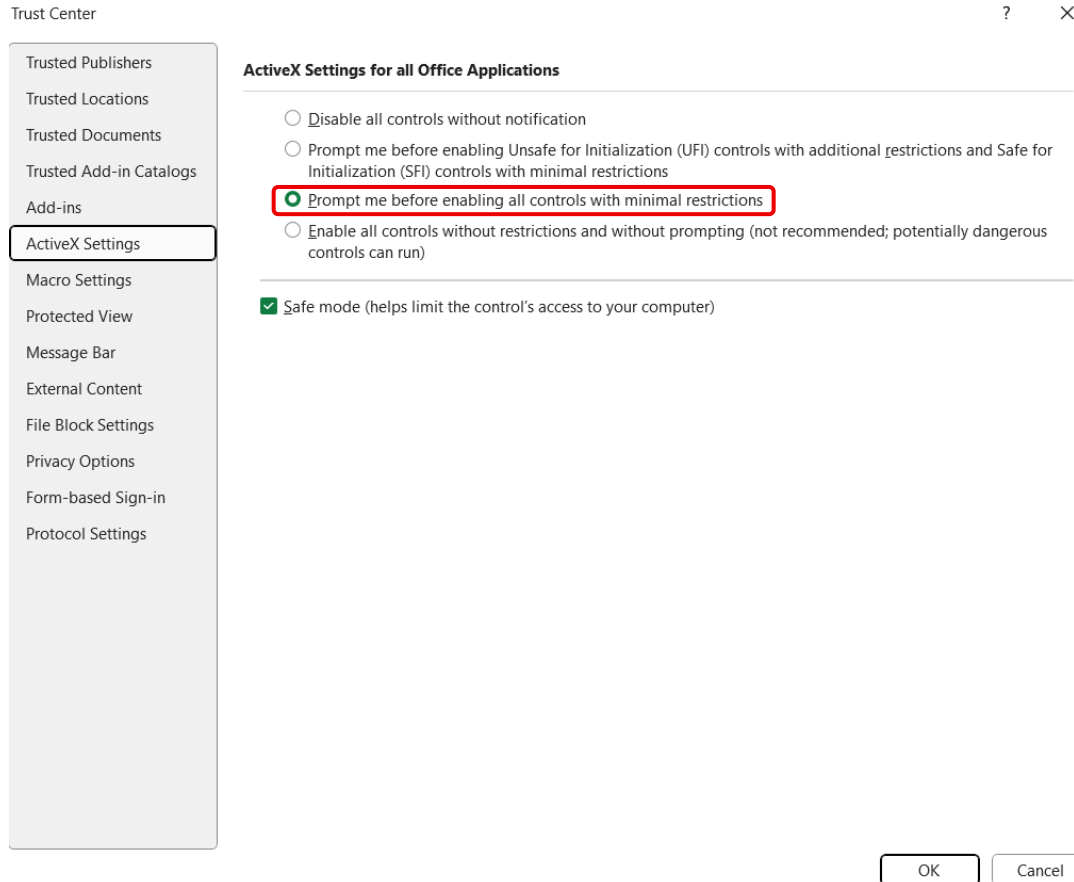
1. Go to **File → Options**
2. Go to **Trust Center → Trust Center Settings**
3. Open **Macro Settings**
4. Select:
 - ✓ *Enable all macros*
 - ✓ *Trust access to the VBA project object model*
5. Click **OK**



8.3 Enable Data Connections

1. Open **Trust Center** again
2. Go to **ActiveX Settings**

3. Enable:



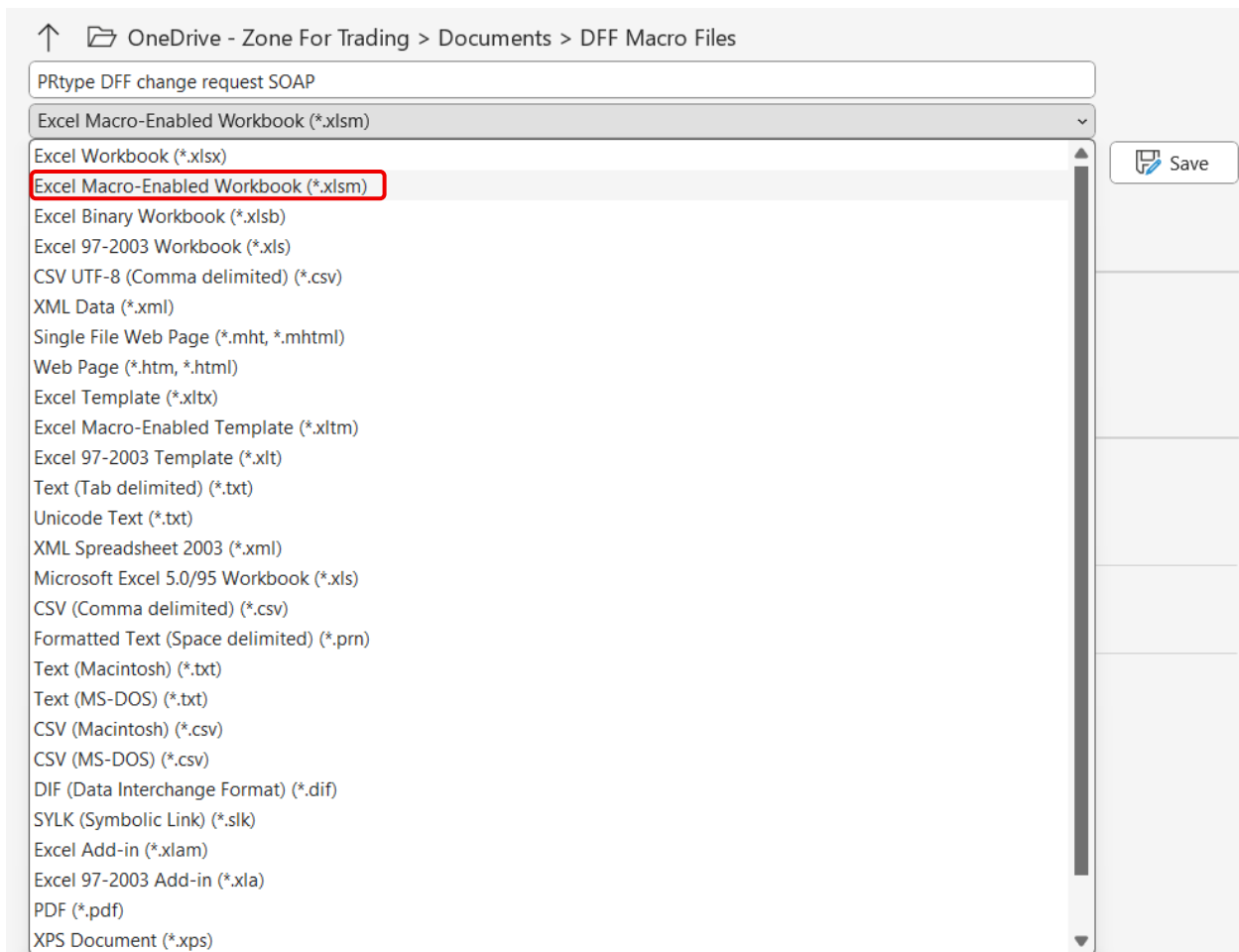
8.4 File Format Requirement

Your Excel file must be saved as:

.xlsm (Macro-Enabled Workbook)

Example file name:

PO_Header_Update_Tool.xlsm



9. VBA Macro Code for Mass Updating PO Headers

Below is the final, optimized macro for the new environment.

✓ Supports:

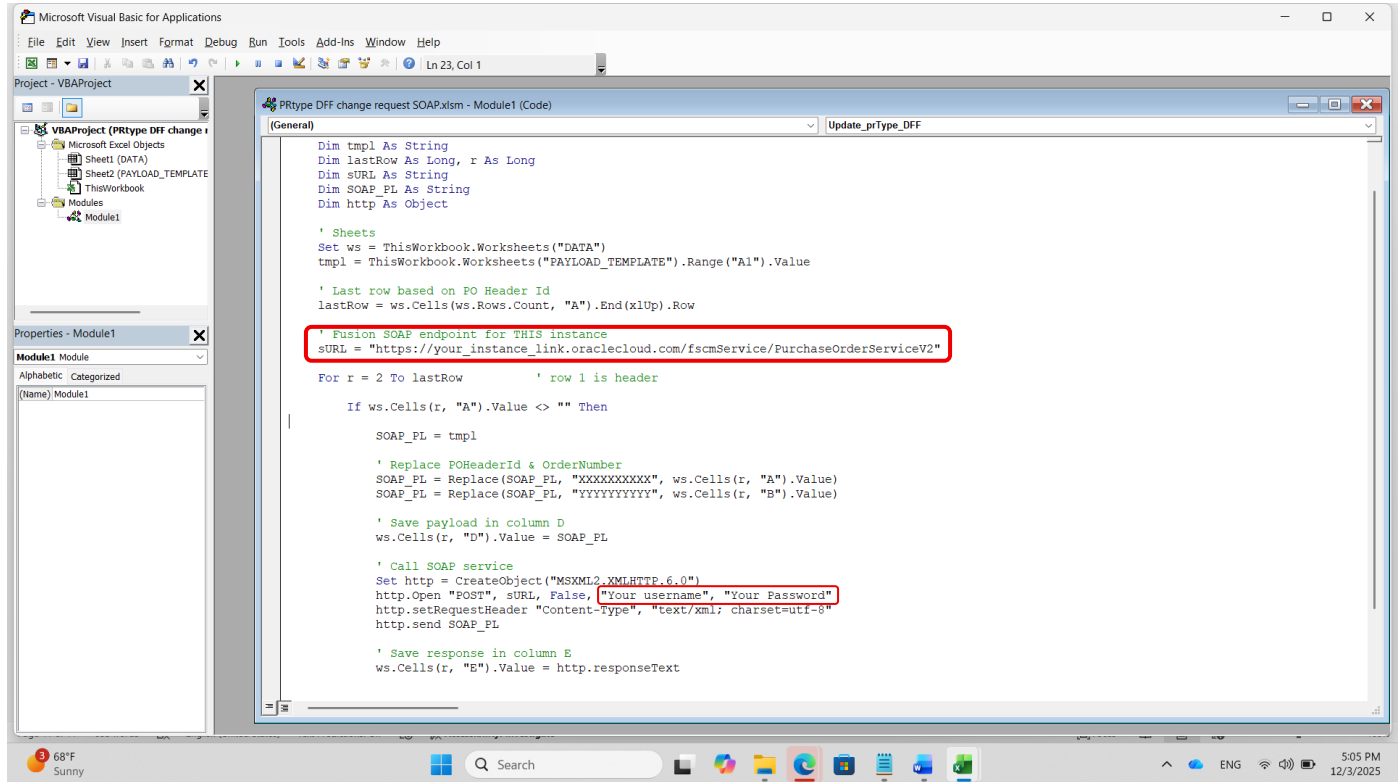
- Standard header attributes
- DFF updates using API name
- Logs payload & response
- Skips blank rows

9.1 VBA Code

ALT + F11

You place this in:

Developer → Visual Basic → Insert Module



10. Running the Update

1. Press **ALT + F8**
2. Select:
 - ✓ Update_prType_DFF
3. Click **Run**
4. Check your DATA Sheet:
 - Column **D** → Payload sent
 - Column **E** → Response